

Classwork 4

Merit and Excellence problem solving

Grades: [A] [M] [E]

Show all working. Finish every proof with a sentence.

Part A · Prove, meet, check

A1 Prove that the sum of two consecutive odd numbers is always a multiple of 4. [M]

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A2 Find the points where $y = x^2 - 3x + 1$ meets $y = x - 2$. [A]

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A3 Solve $\sqrt{x+2} = x$. [M]

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Part B · A tangent and a calendar

B1 Find k so that the line $y = kx - 1$ is a tangent to $y = x^2 + 3$. [M]

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B2 On a calendar, a rectangle 2 days wide and 3 weeks tall has top-left number n . Show that the products of its diagonally opposite corners always differ by 14. [M]

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Part C · A largest area

C A rectangle has perimeter 24 cm. Show that its area cannot be more than 36 cm^2 .

[E]

Words you may use

width x · quadratic · real solution · discriminant · therefore

Sentence starters (optional): *If the width is x , the length is ...* *For a real width, ...*

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Extension

Where do $x^2 + y^2 = 25$ and $y = x + 1$ meet?

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